Project Overview
What problem the database solves, who it is for, and what the final database contains.

For this project I designed and populated a colorectal cancer database. The information in this database is both molecular and clinical, sourced from the Cancer Genome Atlas (TCGA), where large amounts of data from cancer patients and samples are available as downloadable TXTs. This kind of data, spread across multiple files with the possibility of multiple redundancies, can be difficult to work with and combine with other information. This database solves this problem by cleaning and storing all the data into a single schema so that the data can be easily found, interpreted, and compiled. Plus, the database can be more easily queried than multiple different files. This database is for researchers and students who are working with colorectal cancer data from TCGA that want to investigate, store, and understand the available information. And, the database can also handle different cancer types due to its schema. The final database contains 1.6 million+ rows of patient information, tumor samples, unique mutations, gene information, and mRNA expression from 106 samples. Using this database, helpful queries such as most differentially expressed genes by survival outcome, most mutated genes across all samples, sorting samples by cancer stage, and more can be performed.

Data sources
Where the data came from, file types, collection dates if relevant, and any restrictions.

All of the data was downloaded from TCGA Colorectal Adenocarcinoma (COAD) cohort available through cBioPortal. Four TXTs were downloaded containing data collected approximately between 2011 and 2016. TCGA data is publicly available through the NIH Genomic Data Sharing Policy, but restrictions make sure that patients are de-identified.
The downloaded files are listed below:

1. Data_mutations.txt
2. Data_clinical_patient.txt
3. Data_clinical_sample.txt
4. Data_mrna_seq_v2_rsem.txt

Data cleaning decisions
What you changed, why you changed it, what you removed, and how missing values, duplicates, inconsistencies, or formatting problems were handled.

To clean the data, the following steps were taken:

1. String patient and sample IDs were replaced with integer keys for all primary and foreign key columns.
2. Yes/No boolean columns (polyps_history, polyps_present) and Integer 0/1 columns (sequenced, copy_number, etc.) were changed to TRUE/FALSE.

3. The mRNA expression file had genes as rows, samples as columns which was reversed during cleaning.
4. All float NaN values and empty strings were written as SQL NULL.
5. Chromosome values were given integer primary keys; chromosome names are stored as strings (1-22, X, Y).
6. Missing values preserved as SQL NULL to show data that is missing from the source material rather than the database itself
7. Duplicate rows with all fields equal were deleted

The purpose of these changes was to make sure duplicates were not overrepresented, the data maintained consistency, and formatting was compatible with the python script that generates the insert SQL code.

Database design decisions
Why the schema is structured the way it is, what assumptions you made, what business rules you implemented, and why.

The main idea of the database schema is to follow the patient to an encounter to the sample information. There are lookup tables for many different categorical sample attributes. This is to minimize redundant entries (such as cancer type) and to also make the database adaptable to different datasets including different cancer types and repeated encounters. You can see the structure of the database more clearly in the database model diagram in the GitHub repository.

Key assumptions that were made:

- Protein expression is boolean only. The source data flags whether protein/phosphoprotein data exists for a sample. A PROTEIN_EXPRESSION table is included in the schema for future use.
- Mutations are based on their position. A mutation_id encodes gene, chromosome, position, and allele change (such as TP53_17_7577548_C_T), so the same genomic event is represented by one mutation row even if it appears in multiple samples.
- The database is not unique to colon cancer data, and is therefore structured to eventually accommodate more data from different cancer types.

Business rules that were implemented:

- Samples must correspond to an encounter, so it can not have a null encounter ID
- RNA expression links to Gene via gene_id, not hugo_symbol, so expression rows will still be valid even if a gene is renamed.

Normal form and deviations
Explain normalization choices and justify any intentional departure from a normal form.

The database schema is normalized to the maximum extent for almost all tables and relationships. While following the normal form sequence, the columns were made sure to have only atomic values. For example, the Consequence field in Mutation sometimes contained two terms separated by & (missense_variant&splice_region_variant). Next, all columns were made to depend on their primary key. Transitive dependencies were removed for categorical attributes which were made to have their own lookup tables so they still only depend on their primary key. The database is normalized to the fourth normal form, as fifth and beyond are not applicable since they deal with joint dependencies and reducing columns.The only intentional departure from normal form was in the cancer and detailed cancer. While these tables only contain a single row, it was made this way to accommodate additional cancer types in the future and make the database adaptable for more data, rather than having multiple separate databases for each cancer type.

Relationships and constraints
Primary keys, foreign keys, unique constraints, null rules, checks, and referential integrity choices.

Relationship	Type	FK Column	Rule
PATIENT → ENCOUNTER	One-to-many	encounter.patient_id	NOT NULL a patient must exist before an encounter
ENCOUNTER → SAMPLE	One-to-many	sample.encounter_id	NOT NULL a sample must belong to an encounter
ENCOUNTER → INTEGRATED_PHENOTYPE	Many-to-one	encounter.integrated_phenotype_id	Nullable phenotype may not be

			determined
SAMPLE → PRESERVATION / MSI / PATHOLOGY / TUMOR_SITE / ONCOTREE / CANCER / DETAILED_CANCER	Many-to-one	sample.*_id	Nullable not all samples have all annotations
MUTATION → CHROMOSOME	Many-to-one	mutation.chromosome_id	Nullable to handle edge cases
MUTATION_GENE	Many-to-many bridge	mutation_id, gene_id	Composite PK both FKs NOT NULL
SAMPLE_MUTATION	Many-to-many bridge	sample_id, mutation_id	Composite PK both FKs NOT NULL
RNA_EXPRESSION → SAMPLE, GENE	Many-to-one x2	sample_id, gene_id	Both NOT NULL expression must be anchored

Foreign keys to lookup tables are nullable to accommodate missing annotations in the source data. Structural foreign keys (encounter_id on SAMPLE, patient_id on ENCOUNTER) are not allowed to be null.

Data dictionary
Table names, field names, data types, key status, definitions, and notes.

Table	Field	Type	Key	Definition

PATIENT	patient_id	INT	PK	Integer key for each patient
PATIENT	sex	VARCHAR(6)		Biological sex: MALE or FEMALE
INTEGRATED_PHENOTYPE	integrated_phenotype_id	INT	PK	Surrogate key
INTEGRATED_PHENOTYPE	phenotype	VARCHAR(12)		Molecular phenotype subtype: EMT, Epithelial, or Hypermutated
ENCOUNTER	encounter_id	INT	PK	Integer key for each encounter
ENCOUNTER	patient_id	INT	FK	References PATIENT
ENCOUNTER	integrated_phenotype_id	INT	FK	References INTEGRATED_PHENOTYPE
ENCOUNTER	age	INT		Age at diagnosis in years
ENCOUNTER	os_status	VARCHAR(10)		Overall survival: 0:LIVING or 1:DECEASED
ENCOUNTER	os_months	FLOAT		Months of overall survival follow-up
ENCOUNTER	dfs_status	VARCHAR(21)		Disease-free survival status
ENCOUNTER	dfs_months	FLOAT		Months of disease-free survival follow-up
ENCOUNTER	cancer_stage	VARCHAR(9)		AJCC stage: Stage I through Stage IV
ENCOUNTER	pathology_t_stage	VARCHAR(3)		Tumour T-stage (e.g. T3, T4a)

ENCOUNTER	pathology_n_stage	VARCHAR(3)		Lymph node N-stage (e.g. N0, N2b)
ENCOUNTER	polyps_history	BOOLEAN		History of colorectal polyps prior to diagnosis
ENCOUNTER	polyps_present	BOOLEAN		Polyps present at time of diagnosis
ENCOUNTER	mutation_rate	FLOAT		Overall somatic mutation rate
ENCOUNTER	mutation_phenotype	VARCHAR(5)		MSS or MSI-H phenotype call
ENCOUNTER	histology	VARCHAR(12)		Mucinous or Not Mucinous
PRESERVATION	preservation_id	INT	PK	Surrogate key
PRESERVATION	preservation_method	VARCHAR(13)		Tissue preservation method (Frozen Tissue in this dataset)
MSI	msi_id	INT	PK	Surrogate key
MSI	msi_status	VARCHAR(5)		Microsatellite instability status: MSS or MSI-H
PATHOLOGY	pathology_id	INT	PK	Surrogate key
PATHOLOGY	pathology_status	VARCHAR(9)		Pathology result (Malignant in this dataset)
TUMOR_SITE	tumor_site_id	INT	PK	Surrogate key
TUMOR_SITE	tumor_site	VARCHAR(16)		Anatomical site of the primary tumour
ONCOTREE	oncotree_id	INT	PK	Surrogate key

ONCOTREE	oncotree_code	VARCHAR(4)		MSK OncoTree cancer classification code
CANCER	cancer_id	INT	PK	Surrogate key
CANCER	cancer_type	VARCHAR(17)		Broad cancer category
DETAILED_CANCER	detailed_cancer_id	INT	PK	Surrogate key
DETAILED_CANCER	detailed_cancer_type	VARCHAR(20)		Specific cancer subtype
SAMPLE	sample_id	INT	PK	Surrogate key
SAMPLE	encounter_id	INT	FK	References ENCOUNTER
SAMPLE	preservation_id	INT	FK	References PRESERVATION
SAMPLE	msi_id	INT	FK	References MSI
SAMPLE	pathology_id	INT	FK	References PATHOLOGY
SAMPLE	tumor_site_id	INT	FK	References TUMOR_SITE
SAMPLE	oncotree_id	INT	FK	References ONCOTREE
SAMPLE	cancer_id	INT	FK	References CANCER
SAMPLE	detailed_cancer_id	INT	FK	References DETAILED_CANCER
SAMPLE	sequenced	BOOLEAN		Whether whole-exome sequencing was performed

SAMPLE	copy_number	BOOLEAN		Whether copy number variation data exists
SAMPLE	mrna_data	BOOLEAN		Whether mRNA expression data exists
SAMPLE	microrna_data	BOOLEAN		Whether miRNA data exists
SAMPLE	methylation_status	BOOLEAN		Whether methylation data exists
SAMPLE	protein_data	BOOLEAN		Whether protein expression data exists
SAMPLE	phosphoprotein_data	BOOLEAN		Whether phosphoprotein data exists
SAMPLE	tmb_nonsynonymous	FLOAT		Tumour mutation burden (nonsynonymous mutations per Mb)
CHROMOSOME	chromosome_id	INT	PK	Surrogate key
CHROMOSOME	chromosome_name	VARCHAR(2)		Chromosome label: 1-22, X, or Y
GENE	gene_id	INT	PK	Surrogate key
GENE	hugo_symbol	VARCHAR(25)	UNIQUE	HGNC Hugo gene symbol
GENE	entrez_gene_id	INT		NCBI Entrez Gene identifier
GENE	strand	CHAR(1)		Genomic strand: + or -

GENE	gene_type	VARCHAR(50)		Gene biotype (not populated in source data)
GENE	annotation	TEXT		Free-text annotation field for future use
MUTATION	mutation_id	INT	PK	Surrogate key
MUTATION	chromosome_id	INT	FK	References CHROMOSOME
MUTATION	start_position	INT		GRCh37 genomic start coordinate
MUTATION	end_position	INT		GRCh37 genomic end coordinate
MUTATION	consequence	VARCHAR(66)		VEP consequence term(s)
MUTATION	variant_classification	VARCHAR(22)		MAF variant classification (e.g. Missense_Mutation)
MUTATION	variant_type	VARCHAR(3)		SNP, INS, or DEL
MUTATION	reference_allele	VARCHAR(261)		Reference genome allele
MUTATION	tumor_seq_allele1	VARCHAR(261)		Tumour allele 1
MUTATION	tumor_seq_allele2	VARCHAR(156)		Tumour allele 2 (alternate)
MUTATION	hgvsc	VARCHAR(190)		HGVS coding sequence notation
MUTATION	hgvsp	VARCHAR(176)		HGVS protein notation
MUTATION	transcript_id	VARCHAR(15)		Ensembl transcript identifier

MUTATION	validation_status	VARCHAR(50)		Validation status (all NULL in source data)
MUTATION	sequencing_phase	VARCHAR(50)		Sequencing phase (all NULL in source data)
MUTATION_GENE	mutation_id	INT	PK/FK	References MUTATION
MUTATION_GENE	gene_id	INT	PK/FK	References GENE
SAMPLE_MUTATION	sample_id	INT	PK/FK	References SAMPLE
SAMPLE_MUTATION	mutation_id	INT	PK/FK	References MUTATION
SAMPLE_MUTATION	t_ref_count	INT		Tumour reference allele read count (all NULL in source)
SAMPLE_MUTATION	t_alt_count	INT		Tumour alternate allele read count (all NULL in source)
SAMPLE_MUTATION	n_ref_count	INT		Normal reference allele read count (all NULL in source)
SAMPLE_MUTATION	n_alt_count	INT		Normal alternate allele read count (all NULL in source)
SAMPLE_MUTATION	mutation_status	VARCHAR(50)		Somatic/germline call (all NULL in source)

RNA_EXPRESSION	expression_id	INT	PK	Surrogate key
RNA_EXPRESSION	sample_id	INT	FK	References SAMPLE
RNA_EXPRESSION	gene_id	INT	FK	References GENE
RNA_EXPRESSION	expression_value	FLOAT		RSEM-normalised mRNA expression value

Script map
A clear description of what each script does and which file should run before or after another file.

Below are the following scripts that were used to create and populate this database in the order in which it should be run:

1. create_schema.sql
    a. Creates all tables, primary keys, foreign keys, and constraints in the target database
2. generate_inserts.py
    a. Reads all four source TXTs and generates a single SQL file of INSERT statements in FK-safe order
3. Insert_data.sql
    a. Populates all tables in the correct order (lookup tables first, bridge tables last), created by generate_inserts.py

You can create the database before generating the SQL inserts, but insert_data must come after create schema to actually populate the database.

Reproduction instructions
Software needed, packages or dependencies, database engine version, and the exact sequence of commands or actions to rebuild the database.

Software	Version	Purpose

Python	3.12+	Run generate_inserts.py
pandas	2.0+	CSV parsing in generate_inserts.py
MySQL	Any modern version	Execute the SQL scripts

Database rebuild step by step:

1. Create the schema. Connect to your database engine and upload the create_schema.sql file.
2. Put all four TXT source files as the file paths constants at the top of the script.
3. Generate INSERT statements:
   Using python generate_inserts.py, this writes insert_data.sql (~209 MB, 1.67 million lines) to the same directory.
4. Load the data on the command line:
   mysql -u user -p dbname < insert_data.sql or psql -U user -d dbname -f insert_data.sql
5. Verify row counts. Database structure should be as follows:

Table	Action	Rows	Type	Collation	Size	Overhead
CANCER	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_0900_ai_ci	16.0 KiB	-
CHROMOSOME	Browse Structure Search Insert Empty Drop	24	InnoDB	utf8mb4_0900_ai_ci	16.0 KiB	-
DETAILED_CANCER	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_0900_ai_ci	16.0 KiB	-
ENCOUNTER	Browse Structure Search Insert Empty Drop	110	InnoDB	utf8mb4_0900_ai_ci	48.0 KiB	-
GENE	Browse Structure Search Insert Empty Drop	19,365	InnoDB	utf8mb4_0900_ai_ci	1.8 MiB	-
INTEGRATED_PHENOTYPE	Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_0900_ai_ci	16.0 KiB	-
MSI	Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_0900_ai_ci	16.0 KiB	-
MUTATION	Browse Structure Search Insert Empty Drop	~67,131	InnoDB	utf8mb4_0900_ai_ci	11.0 MiB	-
MUTATION_GENE	Browse Structure Search Insert Empty Drop	~72,198	InnoDB	utf8mb4_0900_ai_ci	4.0 MiB	-
ONCOTREE	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_0900_ai_ci	16.0 KiB	-
PATHOLOGY	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_0900_ai_ci	16.0 KiB	-
PATIENT	Browse Structure Search Insert Empty Drop	110	InnoDB	utf8mb4_0900_ai_ci	16.0 KiB	-
PRESERVATION	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_0900_ai_ci	16.0 KiB	-
PROTEIN_EXPRESSION	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_0900_ai_ci	48.0 KiB	-
RNA_EXPRESSION	Browse Structure Search Insert Empty Drop	~1,412,943	InnoDB	utf8mb4_0900_ai_ci	98.7 MiB	-
SAMPLE	Browse Structure Search Insert Empty Drop	110	InnoDB	utf8mb4_0900_ai_ci	144.0 KiB	-
SAMPLE_MUTATION	Browse Structure Search Insert Empty Drop	~74,898	InnoDB	utf8mb4_0900_ai_ci	6.0 MiB	-
TUMOR_SITE	Browse Structure Search Insert Empty Drop	7	InnoDB	utf8mb4_0900_ai_ci	16.0 KiB	-
18 tables	Sum	~1,646,906	InnoDB	utf8mb4_0900_ai_ci	122.0 MiB	0 B

Limitations and future work

Known weaknesses, source-data issues, scope limits, and ideas for improvement.

Something that could be perceived as a weakness in this database schema is that many of the features that are implemented, such as protein expression and lookup tables for cancer types, are not currently being fully utilized since the database currently only contains colon cancer data available in the files. However, these features are to make the database more adaptable and versatile, and to negate the need for more databases for different cancer types in the future. In terms of the source data, there are many empty values, especially in sample_mutation. However, these columns were contained in order to accommodate found data in the future. Similarly, some samples do not have RNA data. In addition, from this source data, each patient only has one encounter, and therefore one sample. This means there is not longitudinal data that would have been very helpful for analysis. While protein expression is a boolean value for its presence or absence, the actual expression is not reported. However, this table is maintained in case this data is found in the future or available for other cancer types. Overall, the source data does have some missing values but the columns are maintained in order to accommodate more data being added as it's generated. Therefore, the scope of analysis that can be done with this database alone is a bit limited. To improve this database, more information should be added such as proteomics data, and other cancer types can be added to make the database a more useful resource to researchers of different areas of study.

A weakness of this project in general is that some files are too big to upload to GitHub, such as the SQL dump and raw mutations file. This means that some data had to be linked to FigShare, which makes the data a bit scattered but the database is still reproducible.